



Technical Interview - Second Round

Overview

Welcome to the second round of our engineering interview process! This round is designed as a technical deep dive where we'll explore your approach to backend architecture and discuss software design decisions together.

Preparation Task: Task Manager Backend

Objective

Build a small backend service for a task management application that demonstrates your preferred approach to architecting backend software. We're interested in seeing **your personal favorite way** of structuring and organizing backend code.

Required Functionality

Please implement a REST API (or GraphQL if you prefer) with the following capabilities:

- **List tasks** - Retrieve all tasks or filtered subsets
- **Create tasks** - Add new tasks to the system
- **Edit tasks** - Update existing task information
- **Delete tasks** - Remove tasks from the system
- **Assign people to tasks** - Associate users/people with specific tasks

What We're Looking For

Quality over Quantity: We're not expecting an extensive feature set. Instead, focus on:

- **Clean Architecture** - Clear separation of concerns, well-defined layers
- **Code Organization** - Logical structure that scales well
- **Design Patterns** - Appropriate use of patterns where they add value
- **Best Practices** - Error handling, validation, documentation
- **Testing Approach** - At least demonstrate your testing strategy (full coverage not required)

Technical Considerations

You're free to use:

- Any architectural pattern you are most comfortable with, but please use NestJS as the backend framework



- Any database solution (in-memory, SQL, NoSQL)
- Any additional libraries or tools that showcase your approach

Please include:

- A README explaining your architectural decisions
- Basic setup instructions
- Any assumptions you've made

Time Investment

We expect this challenge to take approximately **one hour** of focused coding time. If you find yourself going significantly over, please stop and document what you would have added given more time.

The Interview Session

Format

The second round interview will **not** involve live coding. Instead, we'll have an architectural discussion based on:

1. Your Task Manager Solution

- Walk through your implementation
- Explain your architectural choices
- Discuss trade-offs you considered
- Share what you might do differently with more time/resources

2. Your Own Project (Optional but Highly Encouraged)

- Please bring a project you've worked on recently
- Be prepared to walk us through the codebase
- Highlight interesting architectural decisions
- Discuss challenges you faced and how you solved them

Discussion Topics

We'll explore topics such as:

- Architecture patterns and when to use them
- Scaling considerations
- Testing strategies
- Code maintainability and evolution
- Performance optimization approaches
- Technology selection criteria



What to Bring

1. Your completed task manager backend (via GitHub link or similar)
2. Access to a personal project you'd like to discuss (if available)
3. Any questions you have about our engineering practices and architecture

Our Goal

This interview is a two-way conversation. We want to understand how you think about software design and learn from your experiences, while also giving you insight into the types of technical challenges we tackle. We're looking forward to an engaging technical discussion!

If you have any questions about the task or need clarification on any requirements, please don't hesitate to reach out.

Note: Remember, we value clean, maintainable solutions over complex ones. Show us the code you'd be proud to maintain and extend in a production environment.